

PRODUCT MANAGEMENT CASE STUDY

TagAlng

From market research to AI-governed product metrics

How I built TagAlng's investor-metrics pipeline end to end : benchmarking the market, prioritizing what to track, designing AI agents to monitor it, and testing the product those agents watch over.

Market Research

Metric Prioritization

Dashboard Design

AI Agents

Data Governance

Product Testing

Sahana Vijay

AI Analyst Intern · Data & AI Team · Phygtl, Inc. (TagAlng)

What TagAlng Is, and Where I Fit In

WHAT IS TAGALNG

A hyperlocal, identity-based connection app — think “the 1,247 people on your block who already get you.” Type a sentence about yourself, and TagAlng surfaces the fellows within a 5-minute walk who share what defines you: parenting stage, sobriety, sport, heritage, faith.

Lana is TagAlng's AI concierge who captures what a person is looking for or has to share (whether they want to host or join an event) and mediates every chat until both sides say yes. The original persona considered is moms.

Phygtl, Inc. is the parent company; TagAlng is pre-revenue, pre-Series A, and building the metrics story it will bring to investors.

MY END-TO-END SCOPE

- Benchmarked TagAlng against 8 market comparables to see how our metrics stack up.
- Prioritized and categorized the 10 metrics that would actually go in front of investors.
- Designed and built the live investor dashboard (Looker Studio + Supabase).
- Configured 9 AI agents in Amplitude to monitor those metrics and self-triage issues.
- Defined the data-governance rules for how agents escalate and ticket.
- Tested Lana (the product those agents watch) to feed findings back into product iteration.

One Question, Three Gaps to Close

“Are we building the right thing, and would an investor believe it?” Asking this exposed three separate gaps and each became a phase of this project.

1

No external benchmark

We had a set of potential investor metrics but no sense of whether they were the right ones, or how they'd read next to comparable companies.

2

Too many possible metrics

Engagement, growth, trust & safety, virality — dozens of candidate metrics existed across these spheres. Investors needed the vital few, not everything we could measure.

3

Monitoring doesn't scale by hand

We were tracking more signals and metrics than one person could check daily. Something had to monitor them for us. Something autonomous.

The four phases that follow map directly onto these three gaps — research closes gap 1, prioritization closes gap 2, and AI agents close gap 3.

Benchmarking TagAlng Against 8 Market Comparables

Scope: products that measure real-world connection between people who share an identity or interest within a small geography — the same job TagAlng does.

Nextdoor Hyperlocal social <i>Public · NYSE: NXDR</i>	Bumble BFF Friend-matching <i>Public · NASDAQ: BMBL</i>	Peanut Moms' community <i>Private</i>	Mush Moms' meet-up app <i>Acquired by Mumsnet, 2024</i>	Meetup Interest-based groups <i>Multiple owners since 2017</i>	Airbnb Experiences Local experiences <i>Public · NASDAQ: ABNB</i>
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Plus 2 qualitative-only comparisons: Geneva (acquired by Bumble, no published metrics) and Craigslist Community Forums (decades-old, founder has said the company tracks numbers for internal curiosity only — both discussed narratively rather than in the comparison table.

Sourcing discipline: public companies (Nextdoor, Bumble, Airbnb) drawn from SEC filings and investor communications; private companies (Peanut, Mush) drawn from funding-round press coverage and founder interviews — documented per-source so every claim is traceable.

What the Benchmark Revealed

MOST COMMONLY TRACKED

Connection-formation metrics. The concept behind TagAlng metrics like the Magic Number, Connection Velocity, and Friendship Compound is the single most frequent idea across peers — Bumble BFF, Peanut, and Mush all report some version of connections or friendships formed. Mush's “cumulative friendships” is functionally the closest match to TagAlng's North Star.

Active-network / density metrics. Cluster Activation Curve and Block Density at LIVE are the second most common metrics that have a map — Nextdoor, Meetup, and Peanut each report some version of active user or active node/group counts.

GAPS AGAINST PEERS

A monetization metric — expected. Nextdoor (ARPU) and Airbnb (attach rate) both pair growth with monetization. TagAlng's metrics stay entirely engagement-focused — appropriate since we're pre-revenue, but a gap that will need closing eventually.

A supply-side / organizer metric — worth considering. Meetup (organizer count) and Airbnb (host applications) both track the people who create supply. TagAlng has no equivalent for the users who host events, which can be considered as a future metric.

Takeaway: the research didn't just validate and help narrow down our metrics — it directly informed how we categorized them in Phase 2.

Prioritizing 10 Investor Metrics

We cut out 5 metrics from our initial set of investor metrics and honed in on 10. Every final metric was sorted into one of four categories — the same categorization now used across the dashboards, agents, and benchmark reports. Notably, our agents consider a wider variety of metrics including common market metrics measured in Amplitude.

Growth

- Block Density at LIVE
- Block-Unlock Velocity
- Cluster Activation Curve

Engagement

- The Magic Number
- Connection Velocity
- Friendship Compound
- Combine-Filter Lift

Trust & Safety

- Identity-Claim Accuracy
- Safety Reports

Virality

- K-Factor

Why no monetization category: TagAlng hasn't entered a steady monetization phase, which makes an ARPU-style metric incalculable for now — all 10 metrics are deliberately engagement- and trust-first, matching where the product actually is.

Turning 10 Prioritized Metrics Into a Live Dashboard



The actual investor metrics dashboard, live in Looker Studio

Built on

Looker Studio, connected live to a PostgreSQL database via Supabase — the same source of truth engineering and the AI agents both read from.

Design system

Every card is color-coded to its category — gold for Growth, mint for Engagement, coral for Trust & Safety, plum for Virality — so an investor or teammate reads the dashboard's structure at a glance, no legend needed.

From Dashboards Someone Checks to Agents That Act

THE TRAP

We drafted nine potential Amplitude agents to save us from manually checking dashboards every day. But if someone still has to read through everything each agent produces, decide what matters, and write it up for engineering, we haven't really saved any work.

The design goal became: each agent hands off what it finds directly, in a format engineering or backend can act on immediately — with as little human sorting in between as possible.

THE SOLUTION

- 9 Custom Amplitude Agents (accessible to everyone in the company), one per metric family (across investor and market metrics), each running on its own daily/weekly cadence.
- Each agent reuses the same alert thresholds already defined when we first built it — no new judgment calls invented.
- Findings land as tickets directly in Jira, not a screenshot or a link to a chart.
- All nine agents use the same note format, so a ticket reads the same regardless of which agent created it.

The Nine Agents at a Glance

#	Agent	Cadence	Destination	Action
1	Event Funnel Health	Daily	Frontend & Backend	Create a ticket for serious drops
2	UX Frustration Monitor	Weekly	Frontend / Design	Send a summary, no ticket needed
3	Signal & Engagement Quality	Weekly	Product / Lana team	Create a ticket for real swings
4	Market Metrics Weekly Digest	Weekly	Whole team	Send a summary, no ticket needed
5	Active Users Pulse	Daily	Backend & Product	Create a ticket for serious drops
6	Session Quality Monitor	Weekly	Frontend / Engineering	Create a ticket when threshold is hit
7	Retention Health	Weekly	Product / leadership	Create a ticket for serious drops
8	Business Funnel Conversion	Daily	Frontend & Backend	Create a ticket, checking #1's tickets first
9	Business Events Daily Digest	Daily	Backend / Product	Create a ticket when threshold is hit

Gold rows are regular summaries (no ticket) — they inform rather than interrupt. All others create tickets automatically when their existing threshold is crossed.

Designing How the Agents Govern Themselves

Self-triage by severity

Each finding is auto-labeled using thresholds already built into the agent: urgent (well past threshold), needs attention soon (just crossed it), or worth watching (small, recurring, no single week alarming).

One consistent format

Every ticket states what changed, which metric or event it affects, how urgent it is, and a starting point for investigating — identical shape no matter which of the 9 agents created it.

Dedupe across overlapping agents

Event Funnel Health and Business Funnel Conversion watch nearly the same journey. The second agent checks for an existing ticket first and adds a note instead of duplicating — turning overlap into a stronger signal.

Human-in-the-loop, not human-in-the-way

Someone glances over what the agents created every week or two — not to rebuild anything, just to catch a bad call or a noisy agent and adjust its instructions. A check-in, not a rebuild.

Note: I was responsible for designing and configuring how the AI Agents work but was not responsible for monitoring its ticketing system or doing any backend setup it required.

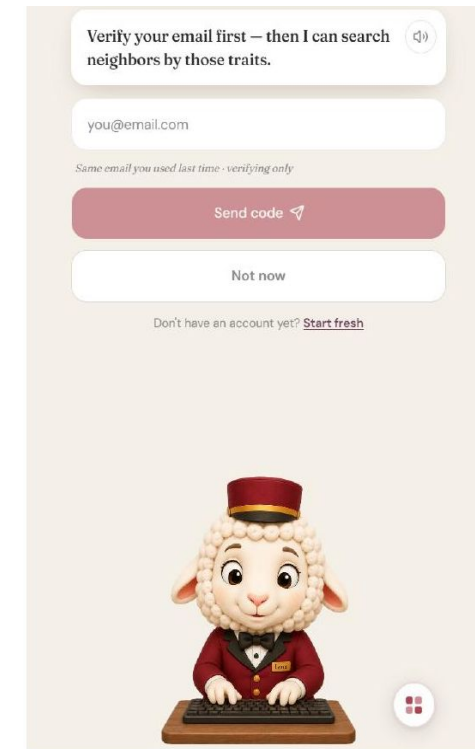
Testing the Product the Agents Watch Over

TEST GOALS

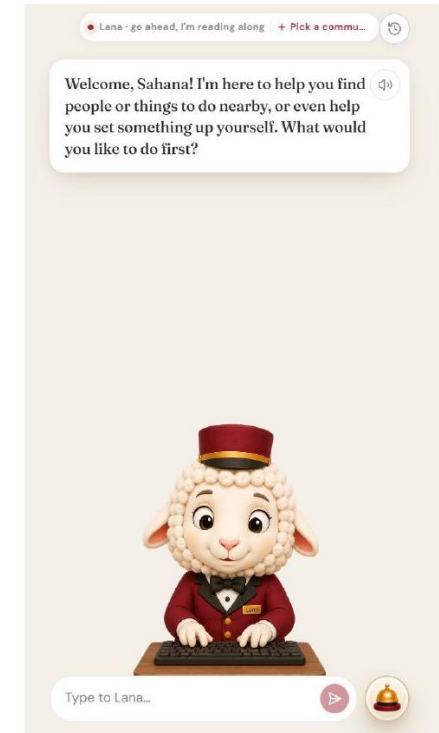
- Test the latest Lana version periodically to uncover conversation & rapport-building problems
- Focus on Day-0 users with no established block — how does the AI build trust from zero?
- Assess design interface, tone, language, and character of the Agent to push toward a truly conversational AI concierge.

For example, in my most recent testing, I role-played a first-time user — a new grad who enjoys reading, looking for people like her — and walked the flow end to end, logging every friction point verbatim as it happened.

Why this connects back to Phase 3: the UX Frustration Monitor and Signal & Engagement Quality agents are built to catch exactly these kinds of rapport-building breakdowns at scale — this manual pass validated what those agents should be watching for.



Email-gate, before any value shown by the agent



Generic welcome, no new-user acknowledgement

Findings → Product Iteration Recommendations

Context gets dropped

Lana re-asks what a user is looking for right after verifying email, even though it's already in her profile.

Recommendation: acknowledge it — “I remember you were searching for X — want to change it?”

Dead-end matching

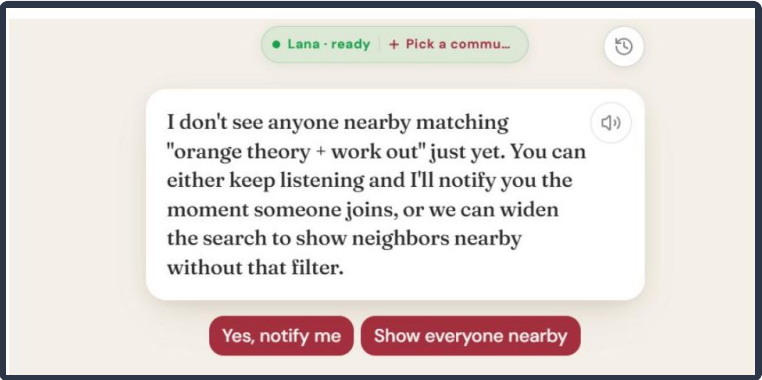
When no one nearby matches, Lana just says so — repeatedly, with no path forward.

Recommendation: after a few failed tries, proactively offer to open a new block for that ZIP.

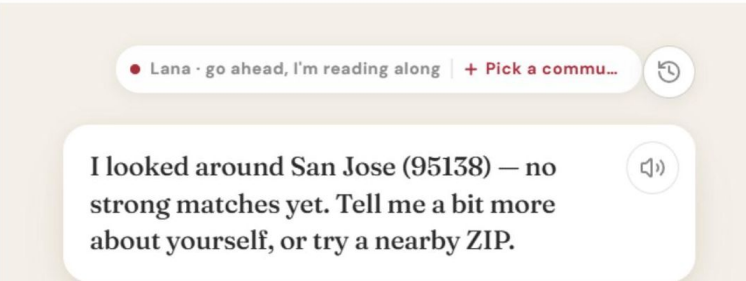
Persona locked to “mom”

The clearest bug: “Graduated university” (education) got filed under “Parenting stage,” and the profile assumed motherhood without it ever being stated.

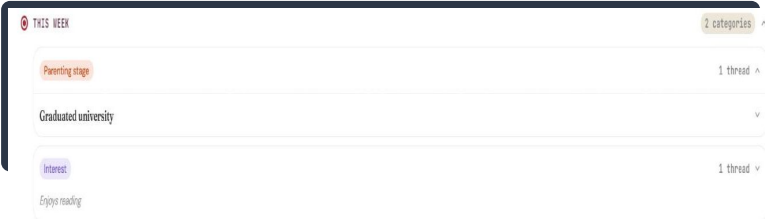
Recommendation: generalize the underlying template beyond the original mom persona.



New interest (OTF) never saved to the profile automatically



“No strong matches” — no offer to open a new block



Actual profile screenshot: education miscategorized as parenting

What This End-to-End Pipeline Proved



Research-grounded metrics

TagAlng's 10 investor metrics are no longer just internal opinion, they're benchmarked against 8 real comparables, with named gaps and named strengths.



A dashboard investors can trust

One live source of truth for investors(Looker + Supabase), color-coded by the same categories used in the research and the agents — no translation needed between artifacts.



Monitoring that scales past one person

9 agents watch what one analyst can't watch alone, self-triage by severity, and only interrupt engineering when a threshold is actually crossed.



A closed loop back to product

Manual testing of Lana didn't just log bugs, it validated what the UX Frustration and Signal Quality agents should flag automatically going forward.

What's Next

ADOPT GROWTH-PLAN FEATURES

- Auto-updating behavioral cohorts — live segments for both agents and marketing
- Custom events & formulas — build combined metrics (e.g. activation rate) natively
- Heatmaps — check whether a flagged drop-off is a discoverability problem first
- Alerts & monitoring — fire the moment a threshold is crossed, not on next agent run
- SSO & project permissions — needed as more tickets land on other teams' boards

EXTEND THE GOVERNANCE LOOP

- Give all 9 agents permission to act on their own (not just make suggestions), with Jira/Linear connections and instructions for what a ticket should say.
- Run another round of Lana testing after the profile-generalization and dead-end fixes ship, to confirm the fixes actually landed.
- Consider a supply-side / organizer metric as an 11th investor metric, per the benchmark's identified gap.
- Keep the biweekly human check-in on agent output as the permanent governance cadence, not a temporary step.

My Contribution

Sahana Vijay

AI Analyst Intern · Data & AI Team

- Competitive market research & benchmark reports.
- Investor-metric prioritization & categorization.
- Looker Studio dashboard designs (Supabase-connected)
- AI agent architecture for 9 Amplitude Custom Agents.
- Data-governance rules: triage, format, dedup logic.
- Product (Lana) testing & iteration recommendations.

Deliverables

TagAlng Investor Metrics Benchmark

Competitive research report, 8 comparables

Live Investor Dashboard

Looker Studio + Supabase, brand design system

Amplitude Agent Usage Plan

9-agent architecture & governance framework

Lana Conversation Testing Notes

Day-0 user testing log with iteration recommendations